RAJALAKSHMI ENGINEERING COLLEGE

Dhanusheya G 251801053-2025-AIDS _ C2

Started on	Thursday, 27 August 2026, 8:06 AM
State	Finished
Completed on	Thursday, 17 September 2026, 9:40 AM
Time taken	21 days 1 hour
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array of N integer, we have to maximize the sum of arr[i] * i, where i is the index of the element (i = 0, 1, 2, ..., N).Write an algorithm based on Greedy technique with a Complexity O(nlogn).

Input Format:

First line specifies the number of elements-n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer: (penalty regime: 0 %)

```c

#include <stdio.h>
#include <stdlib.h>

int compare(const void *a, const void *b) {
    long long x = *(const long long *)a;
    long long y = *(const long long *)b;

    if (x < y)
        return -1;
    if (x > y)
        return 1;
    return 0;
}

int main() {
    int n;
    scanf("%d", &n);

    long long arr[n];

    for (int i = 0; i < n; i++) {
        scanf("%lld", &arr[i]);
    }

    // Greedy: sort in ascending order
    qsort(arr, n, sizeof(long long), compare);

    long long sum = 0;

    for (int i = 0; i < n; i++) {
        sum += arr[i] * i;
    }

    printf("%lld\n", sum);

    return 0;
}

```

	Input	Expected	Got	
✔	5 2 5 3 4 0	40	40	✔
✔	10 2 2 2 4 4 3 3 5 5 5	191	191	✔
✔	2 45 3	45	45	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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